Supporting Information

Probing Generative Protein Structure Models with Transition Path Sampling

Muhammad R. Hasyim
Simons Center for Computational Physical Chemistry, New York University, New York, NY 10003

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1 Extended State Space and Notation

1.1 Boltz-2 reverse sampler as discrete-time stochastic dynamics

Boltz-2 [1] generates biomolecular structures via reverse diffusion: starting from Gaussian noise, the model iteratively denoises atom coordinates over L steps until a plausible 3-D structure emerges. We treat this reverse sampler as a *discrete-time, inhomogeneous Markov chain* and apply the Bolhuis–Chandler–Dellago transition path sampling (TPS) [2, 3] framework to it.

At each diffusion step $i \in \{0, 1, \ldots, L\}$, define the extended state

$$\mathcal{S}_i = \big(x_i,\ R_i,\ \tau_i,\ \varepsilon_i,\ i\big), \tag{1}$$

where:

- $x_i \in \mathbb{R}^{M\times 3}$: atom coordinates for M atoms.
- $R_i \in \mathrm{SO}(3)$: random rotation drawn from the Haar (uniform) measure, implemented via `random_quaternions`.
- $\tau_i \in \mathbb{R}^{1\times 3}$: translation from $\mathcal{N}(0, s_{\text{trans}}^2 I)$, $s_{\text{trans}} = 1.0$.
- $\varepsilon_i \in \mathbb{R}^{M\times 3}$: "churn" noise at step i.

The TPS boundary states are:

$$\text{State } A : x_0 \sim \mathcal{N}(0,\ \sigma_{\max}^2 I) \quad \text{(pure noise)}, \tag{2}$$

$$\text{State } B : \phi(x_L) \in B \quad \text{(reactive endpoint, e.g. pLDDT threshold or RMSD cutoff)}. \tag{3}$$

SE(3) augmentation (Option A). At each step Boltz-2 applies a random rigid transformation to the current coordinates before noise injection and denoising. Including (R_i, τ_i) in the extended state preserves the exact path probability factorization. Crucially, the Haar and translation terms cancel exactly in all acceptance ratios (proved in section 4.3), so the only runtime overhead is storing two small tensors per step.

SE(3) augmentation index (implementation note). When `single_backward_step`(x_{i+1}, i) runs, it samples a fresh augmentation (R_i, τ_i) and stores it on the *result* frame x_i. For a backward-generated frame x_i, the correct augmentation to pass to `recover_forward_noise` is $s_0.R$ and $s_0.\tau$, not $s_1.R$ and $s_1.\tau$.

2 Full Transition Kernel with SE(3) Augmentation

2.1 The one-step map

Each reverse-diffusion step consists of three sub-steps:

(a) SE(3) augmentation.

$$\tilde{x}_i = R_i \cdot \big(x_i - \bar{x}_i\big) + \tau_i, \qquad R_i \sim \mu_{\text{Haar}}\big(\mathrm{SO}(3)\big), \quad \tau_i \sim \mathcal{N}\big(0,\ s_{\text{trans}}^2 I\big), \tag{4}$$

where $\bar{x}_i = \frac{1}{M}\sum_m (x_i)_m$ is the centroid.

(b) Churn noise injection. Define $\hat{t}_i = \sigma_{i-1}(1+\gamma_i)$ and inject

$$x_i^{\text{noisy}} = \tilde{x}_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}\big(0,\ v_i\, I\big), \quad v_i = \eta^2\big(\hat{t}_i^2 - \sigma_{i-1}^2\big). \tag{5}$$

When $\gamma_i = 0$, $v_i = 0$ and no noise is injected.

(c) Deterministic denoising update.

$$x_{i+1} = (1+\alpha_i)\, x_i^{\text{noisy}} - \alpha_i\, D_\theta(x_i^{\text{noisy}},\, \hat{t}_i), \tag{6}$$

where $\alpha_i := s(\sigma_i - \hat{t}_i)/\hat{t}_i$ and D_θ is the EDM-preconditioned denoiser (Section 3.2).

2.2 The transition kernel

$$p\big((x_{i+1}, R_i, \tau_i, \varepsilon_i) \mid x_i\big) = \mu_{\text{Haar}}(R_i) \cdot \mathcal{N}(\tau_i;\, 0,\, s_{\text{trans}}^2 I) \cdot \mathcal{N}(\varepsilon_i;\, 0,\, v_i I) \cdot \delta\big(x_{i+1} - T_i(x_i; R_i, \tau_i, \varepsilon_i)\big). \tag{7}$$

2.3 Backward step: SE(3) and fixed-point inversions

SE(3) inversion.

$$x_i = R_i^\top (\tilde{x}_i - \tau_i) + \bar{x}_i. \tag{8}$$

Since R_i is stored in the snapshot, this is exact and $\mathcal{O}(M)$.

Fixed-point inversion for x_i^{noisy}. The forward update satisfies $x_{i+1} = (1+\alpha)\, x^{\text{noisy}} - \alpha\, D_\theta(x^{\text{noisy}}, \hat{t})$. The naive fixed-point iteration diverges at low noise. The rescaled iteration

$$x \;\leftarrow\; \frac{x_{i+1} + \alpha\big(\hat{x}(x, \hat{t}) - c_{\text{skip}}\, x\big)}{\gamma_{\text{eff}}}, \qquad \gamma_{\text{eff}} = 1 + \alpha(1 - c_{\text{skip}}), \tag{9}$$

is contractive and converges in four iterations in practice.

3 Boltz-2 as Karras Stochastic Sampler

3.1 VESDE forward process

Following Song et al. [5], the forward process is a Variance-Exploding SDE (VESDE):

$$p(x_t \mid x_0) = \mathcal{N}\big(x_t;\, x_0,\, \sigma(t)^2 I\big). \tag{10}$$

The reverse process is steered by the score $\nabla_x \log p_\sigma(x) = (D_\theta(x,\sigma) - x)/\sigma^2$ (Tweedie's formula).

3.2 EDM preconditioning

Karras et al. [4] introduce preconditioning:

$$D_\theta(x, \sigma) = c_{\text{skip}}(\sigma)\, x + c_{\text{out}}(\sigma)\, f_\theta\big(c_{\text{in}}(\sigma)\, x,\; c_{\text{noise}}(\sigma)\big), \tag{11}$$

with scalars

$$c_{\text{skip}}(\sigma) = \frac{\sigma_{\text{d}}^2}{\sigma^2 + \sigma_{\text{d}}^2}, \qquad c_{\text{out}}(\sigma) = \frac{\sigma\, \sigma_{\text{d}}}{\sqrt{\sigma^2 + \sigma_{\text{d}}^2}}, \tag{12}$$

$$c_{\text{in}}(\sigma) = \frac{1}{\sqrt{\sigma^2 + \sigma_{\text{d}}^2}}, \qquad c_{\text{noise}}(\sigma) = \tfrac{1}{4} \ln\big(\sigma/\sigma_{\text{d}}\big). \tag{13}$$

3.3 Churn as a finite Langevin correction

Remark 3.1. *For $\gamma_i = 0.8$ (Boltz-2 default), the churn noise variance is $\eta^2 \gamma_i (2+\gamma_i) \sigma_{i-1}^2$, a* finite *amount independent of the step size $|\sigma_i - \sigma_{i-1}|$. This is* not *an Euler–Maruyama discretisation of a fixed SDE; as $\gamma_i \to 0$, stochasticity vanishes and the sampler reduces to the deterministic probability flow ODE.*

4 Path Probability and Jacobians

4.1 Log path probability

$$\log P(\mathcal{X}) = \log \rho(x_0) + \sum_{i=0}^{L-1} \Big[\underbrace{\log \mu_{\text{Haar}}(R_i)}_{\text{const.}} + \underbrace{\log \mathcal{N}(\tau_i;\, 0,\, s_{\text{trans}}^2 I)}_{\text{Gaussian}} + \underbrace{\log \mathcal{N}(\varepsilon_i;\, 0,\, v_i I)}_{\text{Gaussian}} - \underbrace{\log |\det J_i|}_{\text{Jacobian}} \Big], \tag{14}$$

where $J_i = \partial x_{i+1} / \partial \varepsilon_i$.

4.2 Exact Jacobian

From Eq. (6):

$$\boxed{J_i = \beta_i\, I + \mu_i\, \nabla_z f_\theta \Big|_{z = c_{\text{in}}(\hat{t}_i)\, x_i^{\text{noisy}}},} \tag{15}$$

with $\beta_i = 1 + \alpha_i(1 - c_{\text{skip}}(\hat{t}_i))$ and $\mu_i = -\alpha_i\, c_{\text{out}}(\hat{t}_i)\, c_{\text{in}}(\hat{t}_i)$.

4.3 Cancellation in acceptance ratios

Proposition 4.1 (Forward shooting: Jacobians cancel)**.** *For one-way forward shooting from point k, the Jacobian log-determinants cancel exactly in the Metropolis acceptance ratio. The remaining acceptance factor is $H_{AB}(X^{\text{n}})$.*

Proof. Fresh $(R_i^{\text{n}}, \tau_i^{\text{n}}, \varepsilon_i^{\text{n}})$ are drawn from their priors for $i \geq k$. The proposal probability $G(X^{\text{n}}|X^{\text{o}}, k) = \prod_{i \geq k} \mu_{\text{Haar}}(R_i^{\text{n}}) \mathcal{N}(\tau_i^{\text{n}}) \mathcal{N}(\varepsilon_i^{\text{n}})$. In the MH ratio $P(X^{\text{n}})/P(X^{\text{o}}) \times G(X^{\text{o}})/G(X^{\text{n}})$, the Haar, translation, and noise terms from G cancel the same factors from P. The Jacobian terms $|\det J_i|$ remain; because α_i depends only on schedule step i, we have $|\det J_i^{\text{n}}| = |\det J_i^{\text{o}}|$ for each $i \geq k$, giving exact cancellation. □

4.4 Two-way acceptance ratio

$$p_{\text{acc}} = H_{AB}(X^{\text{n}}) \cdot \min\left[1,\ \frac{\rho(x_0^{\text{n}})}{\rho(x_0^{\text{o}})} \prod_{i=0}^{k-1} \frac{p_{\text{fwd}}(x_{i+1}^{\text{n}} \mid x_i^{\text{n}})\, p_{\text{bwd}}(x_i^{\text{o}} \mid x_{i+1}^{\text{o}})}{p_{\text{fwd}}(x_{i+1}^{\text{o}} \mid x_i^{\text{o}})\, p_{\text{bwd}}(x_i^{\text{n}} \mid x_{i+1}^{\text{n}})}\right], \tag{16}$$

where

$$\log p_{\text{fwd}}(x_{i+1} \mid x_i) = -\frac{\|\varepsilon_i^*\|^2}{2v_i} - \frac{3M}{2} \log(2\pi v_i) - \log |\det J_i|, \tag{17}$$

$$\log p_{\text{bwd}}(x_i \mid x_{i+1}) = -\frac{\|x_i - x_{i+1}\|^2}{2(\sigma_{i-1}^2 - \sigma_i^2)} - \frac{3M}{2} \log\big(2\pi(\sigma_{i-1}^2 - \sigma_i^2)\big). \tag{18}$$

In implementation form:

$$\log r = (\text{lf_new} - \text{lf_old}) + (\text{lb_old} - \text{lb_new}) + \big(\log \rho(x_0^{\text{n}}) - \log \rho(x_0^{\text{o}})\big). \tag{19}$$

5 TPS in the Extended Ensemble

5.1 Fixed-length reactive path ensemble

Following Falkner et al. [3], the extended ensemble is

$$\pi(\mathcal{X}, \hat{k}) \propto P_{AB}(\mathcal{X})\, P(\mathcal{X})\, p(\hat{k} \mid \mathcal{X}), \tag{20}$$

where $P_{AB}(\mathcal{X}) = \mathbb{1}[x_0 \in A,\, x_L \in B]$.

5.2 Move set and ergodicity

The complete move scheme combines three moves:

- **45% forward shooting** (uniform selector over $\{1, \ldots, L-1\}$; always accepts reactive paths, $p_{\text{acc}} = H_{AB}(X^{\text{n}})$).
- **45% backward shooting** (generates new prefix via q_{bwd}; MH-corrected via Eq. (19)).
- **10% global reshuffle** (draws fresh $x_0 \sim \rho$ and all ξ_i; always accepts, $p_{\text{acc}} = H_{AB}(X^{\text{n}})$).

Theorem 5.1 (Ergodicity)**.** *The Markov chain on the reactive path ensemble $\mathcal{R}$ defined by the above move set is ergodic with stationary distribution π, without requiring connectivity assumptions on path space.*

Proof sketch. Forward + backward shooting jointly update both endpoints and satisfy detailed balance. The reshuffle reaches any reactive path in a single step, guaranteeing irreducibility. □

NaN guard. The Boltz-2 model can produce NaN outputs in degenerate configurations. Four defence layers are implemented: (i) `weighted_rigid_align` exits early on non-finite input; (ii) `_nan_fallback()` in `gpu_core.py` replaces non-finite values with the previous state; (iii) `trajectory_has_finite_coords()` sets bias to 0 for NaN trials; (iv) an `isfinite` check in OpenPathSampling [8] prevents NaN probabilities from being accepted.

6 OPES-Enhanced TPS Acceptance

6.1 On-the-fly Probability Enhanced Sampling

The OPES method [6, 7] builds an adaptive bias $V_n(\mathbf{s})$ on the CV $\mathbf{s} = \mathbf{s}(x_L)$ via on-the-fly kernel density estimation with compression. The KDE estimate is

$$P_n(\mathbf{s}) = \frac{1}{W} \sum_{k=1}^{n_k} h_k \prod_{d=1}^{D} G\big(s^d;\, c_k^d,\, \Sigma_k^d\big), \tag{21}$$

where G is a truncated Gaussian, D is the CV dimension, and $W = \sum_k h_k$ (convergence mode) or n_k (explore mode). The bias is

$$V_n(\mathbf{s}) = (1 - 1/\gamma)\, k_B T \, \log\!\left(\frac{P_n(\mathbf{s})}{Z_n} + \epsilon\right), \tag{22}$$

where γ is the well-tempered biasfactor and ϵ is a small regularizer preventing divergence in unsampled regions.

6.2 Modified TPS acceptance with OPES bias

The OPES bias enters the TPS Metropolis acceptance as a multiplicative factor on the trial weight:

$$p_{\text{acc}}^{\text{OPES}} = H_{AB}(X^{\text{n}}) \cdot \min\!\left[1,\; r_{\text{TPS}} \cdot \exp\!\Big(-\frac{V_n(\mathbf{s}^{\text{n}}) - V_n(\mathbf{s}^{\text{o}})}{k_B T}\Big)\right], \tag{23}$$

where r_{TPS} is the standard TPS log-ratio from Eq. (19) (unity for forward-only shoots), and $\mathbf{s}^{\text{n,o}}$ are the CVs of the new and old accepted paths evaluated at the terminal frame x_L. The bias V_n is updated after every `pace` MC steps by depositing a new kernel centred at the current $\mathbf{s}$ (Eq. (21)).

6.3 Two-dimensional CV case: ligand pose and pocket distance

For the co-folding application, we use the two-dimensional CV $\mathbf{s} = (s_1, s_2)$ where:

- s_1 = ligand pose RMSD (Å): Kabsch-aligned heavy-atom RMSD of the predicted ligand pose to a reference crystal pose.
- s_2 = ligand–pocket distance (Å): mean distance between the ligand center-of-mass and the nearest binding-pocket Cα atoms.

The native binding configuration corresponds to small s_1 and moderate s_2, while alternative poses or displaced ligands appear at larger s_1 or at the tails of the s_2 distribution.

Bias parameters for the case-1 2D run. $\gamma = 10$, barrier $= 5\,k_BT$, pace $= 1$, kernel compression threshold $= 1.0$; 1159 depositions and 82 surviving kernels (after compression) at the end of the run.

7 Collective Variables for Diagnostics

The following CVs are available in `collective_variables.py`. They are evaluated on the current accepted path and logged to `cv_values.json`. *Diagnostic CVs do not feed into the OPES bias.*

Bias CVs (sampling).

Ligand pose RMSD (s_1) Kabsch-aligned heavy-atom RMSD of the predicted ligand to a reference pose. Low values indicate native-like binding; large values flag off-target poses or model hallucinations.

Ligand–pocket distance (s_2) Mean distance from ligand center-of-mass to nearest binding-pocket Cα atoms (Å). Together with s_1 this provides a 2-D fingerprint of pose quality.

OpenMM Cα-RMSD to local minimum (single-chain runs) Each terminal frame is energy-minimised with AMBER14/GBn2 (OpenMM, L-BFGS); Kabsch-aligned Cα-RMSD to the minimised structure. Low values: near-stable fold; large values: hallucinated or disordered.

Geometric diagnostics.

Contact order Fraction of long-range Cα pairs (separation ≥ 6, threshold 8 Å):

$$\mathrm{CO}(x) = \frac{|\{(i,j) : |i-j| \geq 6,\ d_{ij} < 8\,\text{Å}\}|}{|\{(i,j) : |i-j| \geq 6\}|}.$$

Detects mode collapse and topology changes.

Steric clash count $N_{\text{clash}} = |\{(i,j) : |i-j| \geq 3,\ d_{ij} < 3.0\,\text{Å},\ i < j\}|$. Flags unphysical packing.

lDDT to reference Alignment-free local distance difference test [9]. Detects memorisation (lDDT ≈ 1) and local errors.

Ramachandran outlier fraction Fraction of residues outside allowed (ϕ, ψ) regions. Detects unphysical backbone geometry.

Radius of gyration $R_g = \sqrt{\frac{1}{M}\sum_m \|x^m - \bar{x}\|^2}$ (Å). Tracks global compaction changes.

Model self-confidence.

Boltz pLDDT Mean per-residue pLDDT (0–100) from the Boltz-2 confidence head. High pLDDT with large RMSD indicates overconfident hallucination.

8 OPES-TPS 2D Case: Quantitative Summary

The run in `opes_tps_out_case1_2d` consists of 11 590 MC steps with 3 184 accepted moves (acceptance rate 27.5%), using the 2-D CV $\mathbf{s} = (s_1, s_2)$ described above. Table 1 summarises the marginal statistics from `scripts/summarize_opes_2d_cv.py`.

Table 1: Marginal CV statistics for the OPES-TPS 2D run (case 1). "All" = over all 11 590 logged MC steps; "Acc." = restricted to 3 184 accepted steps.

	Ligand RMSD s_1 (Å)			Pocket dist. s_2 (Å)		
Subset	Mean ± SD	Median	p_{95}	Mean ± SD	Median	p_{95}
All	0.49 ± 0.35	0.49	1.19	5.44 ± 0.11	5.40	5.62
Acc.	0.38 ± 0.32	0.26	0.95	5.41 ± 0.10	5.37	5.61

Tail mass. Among all logged steps, 48.9% have $s_1 > 0.5$ Å, 23.5% have $s_1 > 0.75$ Å, 8.8% have $s_1 > 1.0$ Å, and 1.7% have $s_1 > 1.25$ Å. Among accepted steps, the corresponding fractions are 36.6%, 14.7%, 4.5%, and 0.9%, confirming that the bias successfully drives the chain to explore elevated-RMSD configurations while the Metropolis filter still preferentially retains lower-RMSD paths.

Run-length structure. Visits to $s_1 > 0.5$ Å occur in 147 distinct runs with a mean run length of 39 consecutive MC steps (maximum 299 steps) over the full trajectory, indicating that tail exploration is not isolated spikes but sustained excursions. Restricting to accepted steps, the same $s_1 > 0.5$ Å threshold has a mean accepted-step run length of ≈ 8 consecutive accepted paths, confirming genuine metastable dwell in higher-RMSD regions.

2-D peak structure. The 2-D histogram of (s_1, s_2) over all logged steps reveals two dominant density peaks: a primary mode at $(s_1, s_2) \approx (0.07\,\text{Å}, 5.37\,\text{Å})$ corresponding to the native-like binding pose, and secondary peaks at $(s_1, s_2) \approx (0.70\,\text{Å}, 5.55\,\text{Å})$ and $(0.83\,\text{Å}, 5.61\,\text{Å})$ representing metastable high-RMSD / larger-pocket-distance configurations.

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